```
> restart :
>
> # КАТАСТРОФА БАБОЧКА
>
> with(plots) :
>
> eq1 := X^5 + a·X^3 + b·X^2 + c·X + d = 0;
```

$$eq1 := X^5 + X^3\,a + X^2\,b + X\,c + d = 0 \quad (1)$$

```
> eq2 := 5·X^4 + 3·a·X^2 + 2·b·X + c = 0;
```

$$eq2 := 5\,X^4 + 3\,X^2\,a + 2\,X\,b + c = 0 \quad (2)$$

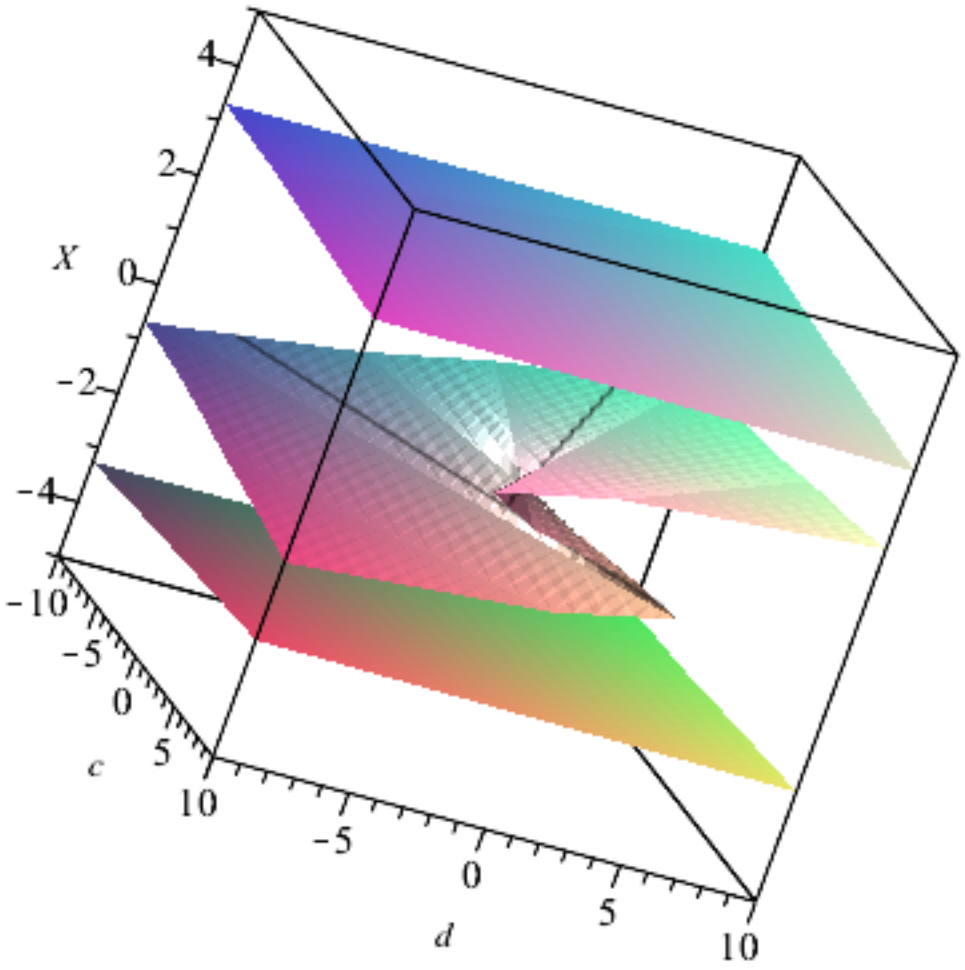
```
>
> eq3 := subs(a = -10, b = 0, eq1);
```

$$eq3 := X^5 - 10\,X^3 + X\,c + d = 0 \quad (3)$$

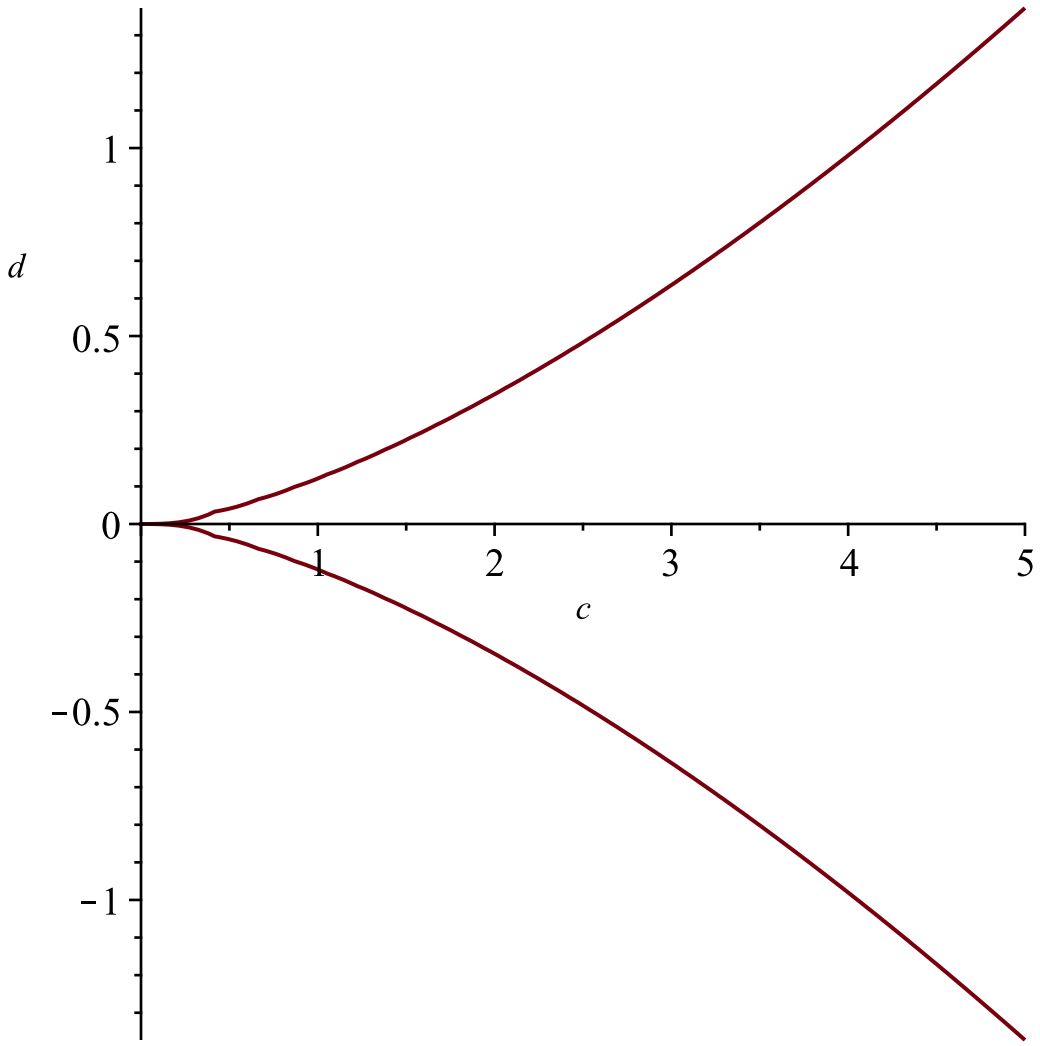
```
> eq4 := subs(a = -10, b = 0, eq2);
```

$$eq4 := 5\,X^4 - 30\,X^2 + c = 0 \quad (4)$$

```
> implicitplot3d(eq3, c = -10 ..10, d = -10 ..10, X = -5 ..5, numpoints = 10000);
```



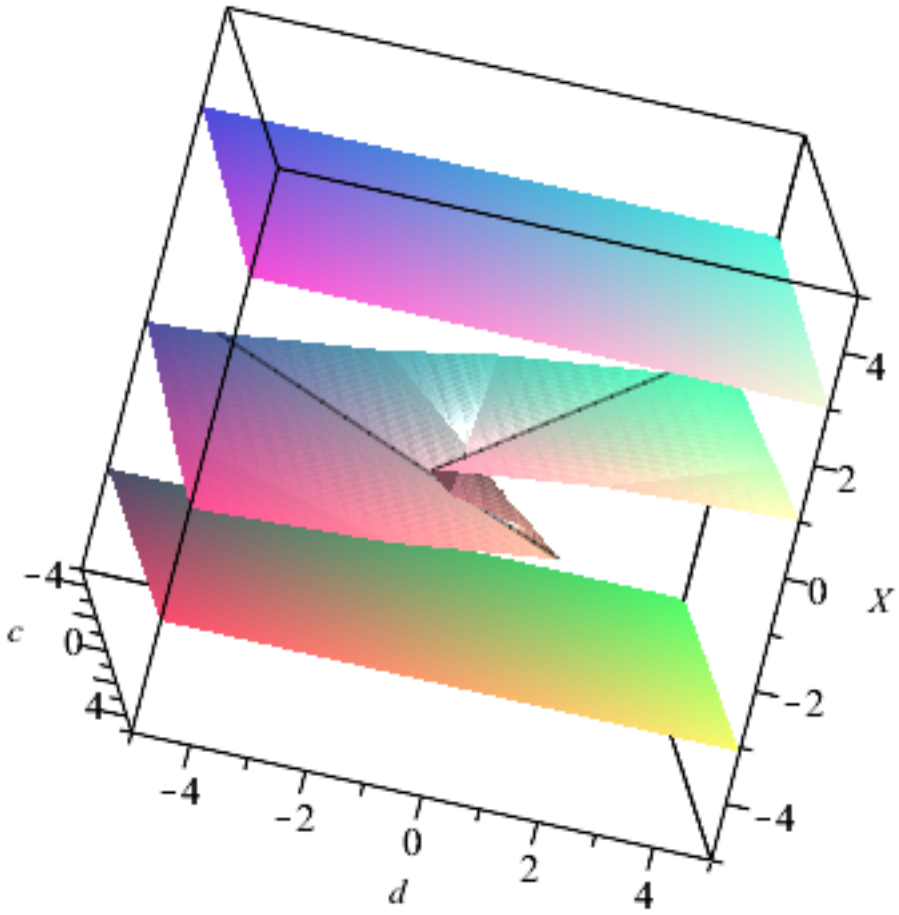

```
> implicitplot([eq3, eq4, X], c = -5 ..5, d = -5 ..5, numpoints = 10000);
```




```
>
> surf_butterfly2 := subs(a = -6, b = 10, eq3);
```

$$surf_butterfly2 := X^5 - 10\,X^3 + X\,c + d = 0 \quad \textbf{(5)}$$

```
> implicitplot3d(eq3, c = -5..5, d = -5..5, X = -5..5, numpoints = 20000);
```




```
> restart :
>
> with(plots) :
>
> eq1 := X^5 + a·X^3 + b·X^2 + c·X + d = 0;
```

$$eq1 := X^5 + X^3\,a + X^2\,b + X\,c + d = 0 \quad (6)$$

```
> eq2 := 5·X^4 + 3·a·X^2 + 2·b·X + c = 0;
```

$$eq2 := 5\,X^4 + 3\,X^2\,a + 2\,X\,b + c = 0 \quad (7)$$

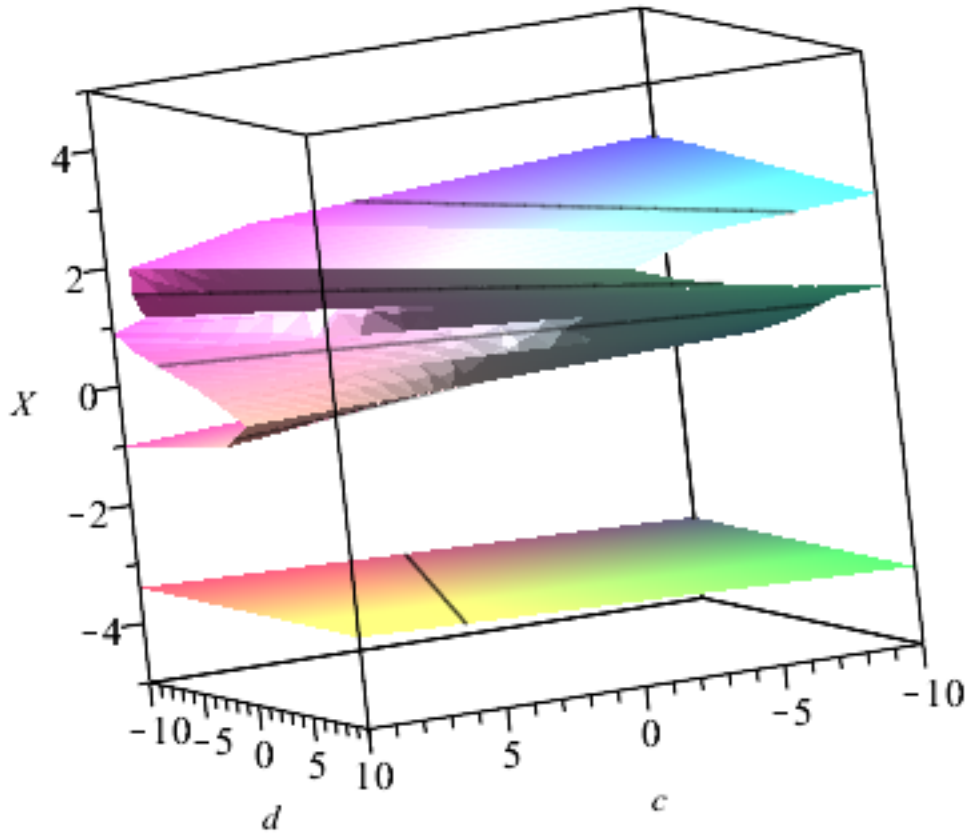
```
>
> eq3 := subs(a = -10, b = 10, eq1);
```

$$eq3 := X^5 - 10\,X^3 + 10\,X^2 + X\,c + d = 0 \quad (8)$$

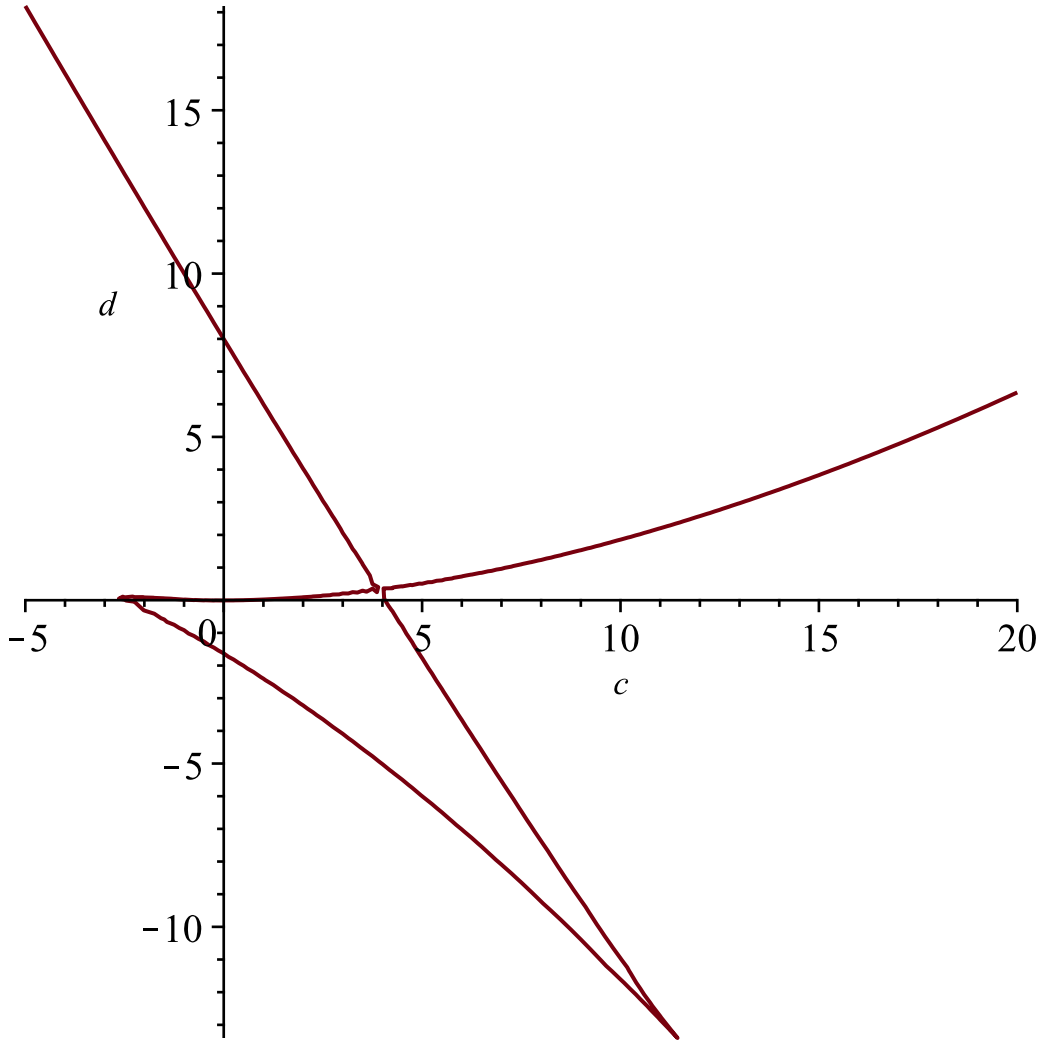
```
> eq4 := subs(a = -10, b = 10, eq2);
```

$$eq4 := 5\,X^4 - 30\,X^2 + 20\,X + c = 0 \quad (9)$$

```
> implicitplot3d(eq3, c = -10..10, d = -10..10, X = -5..5, numpoints = 10000);
```



> *implicitplot*([*eq3*, *eq4*, *X*], *c* = −5 ..20, *d* = −20 ..20, *numpoints* = 10000);




```
> restart :
> with(plots) :
> eq1 := X^5 + a·X^3 + b·X^2 + c·X + d = 0;
```

$$eq1 := X^5 + X^3\,a + X^2\,b + X\,c + d = 0 \quad (10)$$

```
> eq2 := 5·X^4 + 3·a·X^2 + 2·b·X + c = 0;
```

$$eq2 := 5\,X^4 + 3\,X^2\,a + 2\,X\,b + c = 0 \quad (11)$$

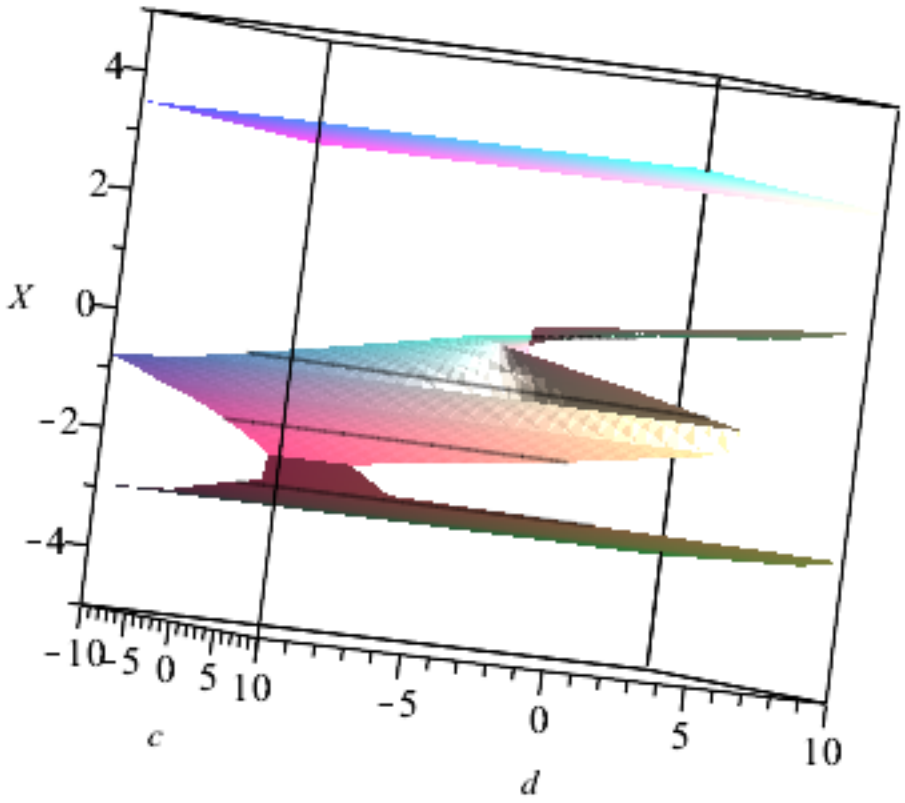
```
> eq3 := subs(a = -10, b = -5, eq1);
```

$$eq3 := X^5 - 10\,X^3 - 5\,X^2 + X\,c + d = 0 \quad (12)$$

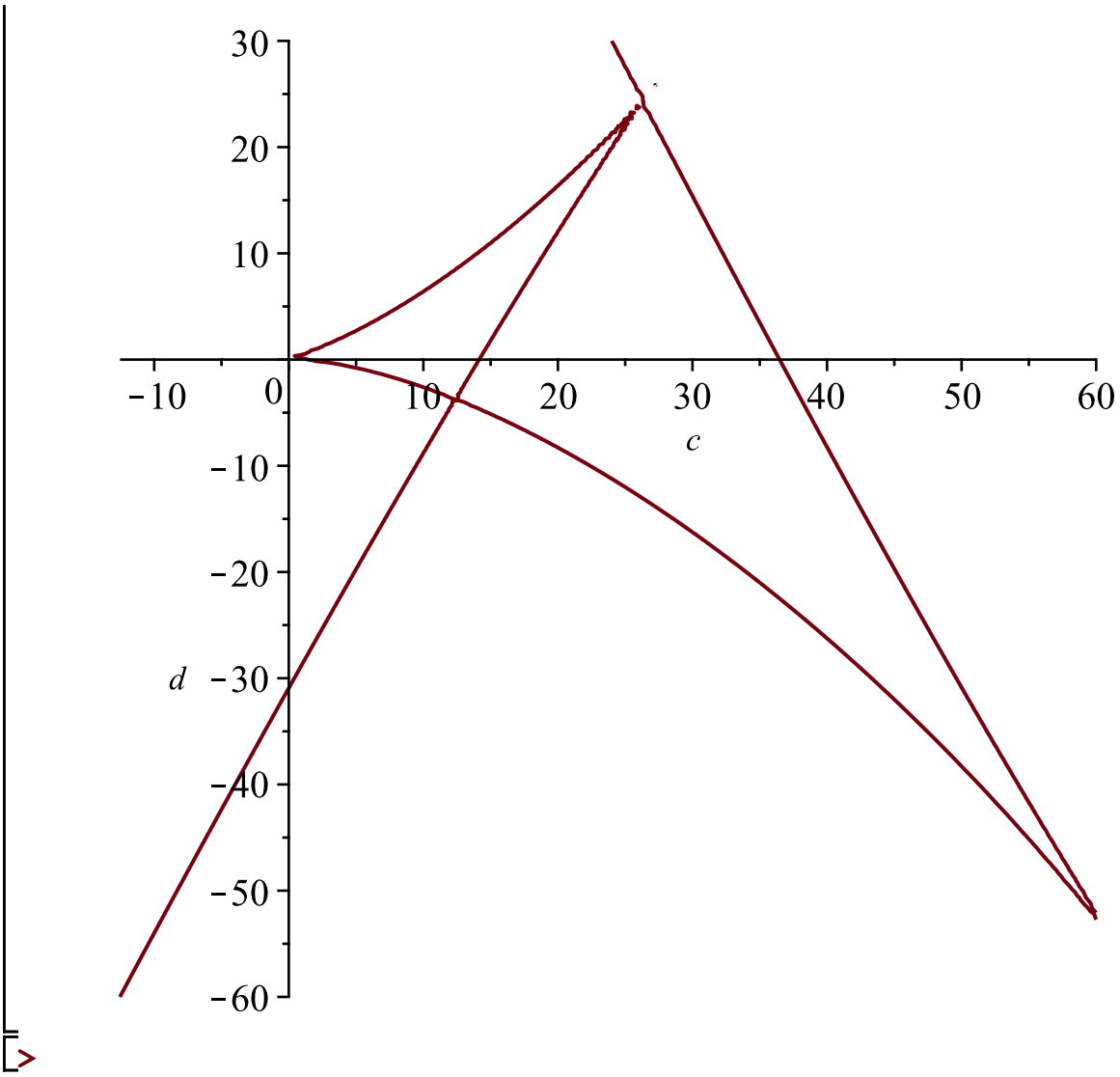
```
> eq4 := subs(a = -10, b = -5, eq2);
```

$$eq4 := 5\,X^4 - 30\,X^2 - 10\,X + c = 0 \quad (13)$$

```
> implicitplot3d(eq3, c = -10..10, d = -10..10, X = -5..5, numpoints = 10000);
```




```
> implicitplot( [eq3, eq4, X], c =-20 ..60, d =-60 ..30, numpoints = 30000 );
```


30
20
10
-10
0
10
20
30
40
50
60
c
-10
-20
d
-30
-40
-50
-60